VLSI Lab

Laboratory report submitted for the partial fulfillment
of the requirements for the degree of

Bachelor of Technology
in
Electronics and Communication Engineering

by

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January 2026

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Experiment - 8

0.1 Aim

1. To design and analyze pass transistor logic using NMOS and PMOS transistors.
2. To observe signal degradation in a 4-stage NMOS and PMOS pass transistor chain.
3. To implement an AND gate using NMOS pass transistor logic.
4. To design a Transmission Gate (TG) using NMOS and PMOS transistors and analyze its benefits.

0.2 Software/ Apparatus/ Components Used

1. LT-SPICE software installed on a computer
2. NMOS transistor model
3. PMOS transistor model
4. Power supply setup (virtual in LT-SPICE)
5. Resistors (values as per circuit design)
6. Signal source (voltage and current sources)

0.3 Theory

Pass transistor logic is a design technique used in low-power digital circuits to implement logic functions using NMOS and PMOS transistors instead of conventional logic gates. Unlike static CMOS logic, PTL reduces transistor count and eliminates redundant switching by allowing signals to pass directly through transistors. However, it introduces challenges like voltage degradation and threshold voltage (Vth) drop in NMOS-based designs. NMOS Pass Transistor: Conducts well for logic LOW (0V) but suffers from Vth drop for logic HIGH (5V). PMOS Pass Transistor: Conducts well for logic HIGH (5V) but has weak logic LOW transmission.
Transmission Gate (TG): A combination of NMOS and PMOS transistors, achieving perfect signal transmission.

0.4 Simulation and Results

0.4.1 Pass Transistor Using NMOS

a) Steps used to draw the Circuit

(a) Connect four NMOS transistors in series, forming a 4-stage pass transistor chain.

(b) Apply 5V input at the first stage and a 5V control signal (Gate) to all NMOS transistors.

(c) Observe the output voltage at each stage using.

b) Schematic Diagram and Simulation and Results

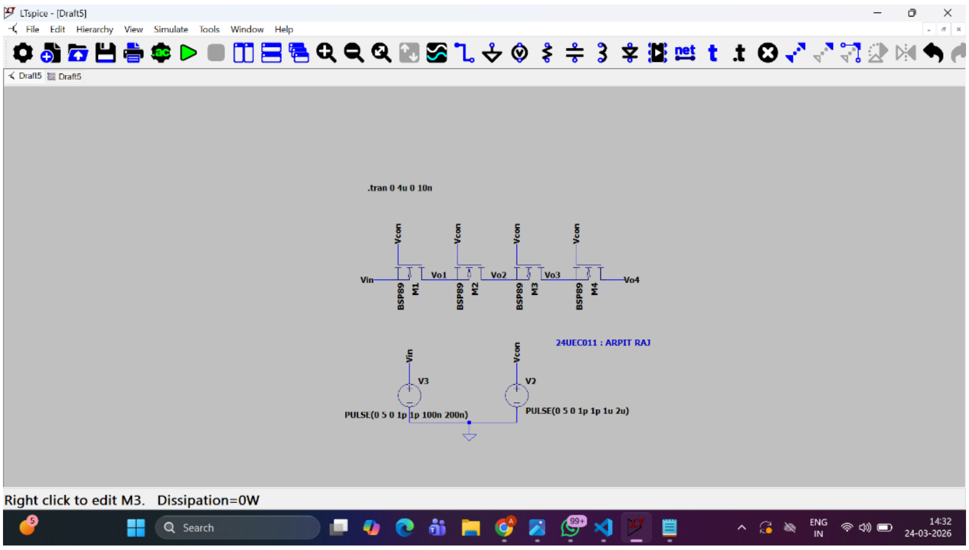

Figure 1: Schematic of Pass Transistor Using NMOS

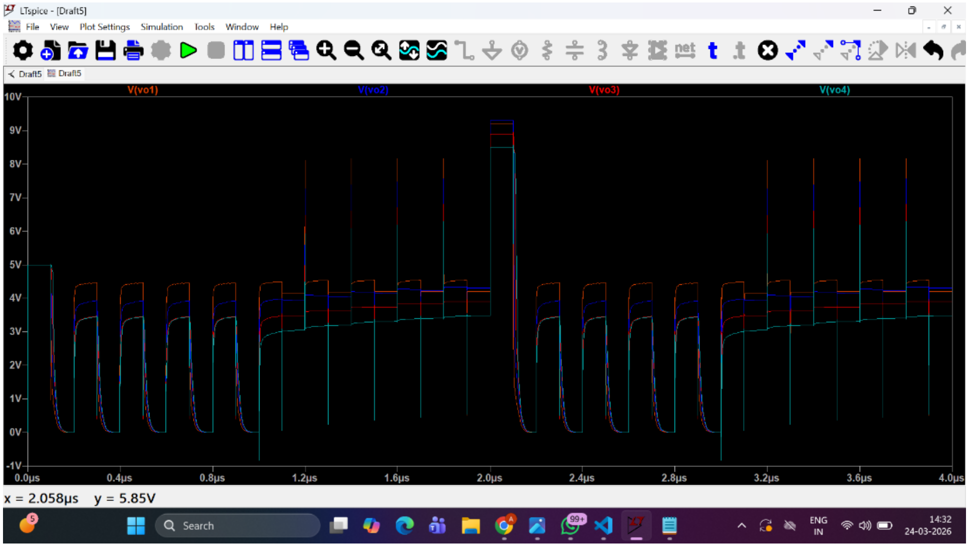

Figure 2: Simulation : Pass Transistor Using NMOS

0.4.2 Pass Transistor Using PMOS

a) Steps used to draw the Circuit

(a) Connect four PMOS transistors in series.

(b) Apply 0V input at the first stage and a 0V control signal to all PMOS transistors.

(c) Observe the output voltage at each stage using a voltage probe.

(d) Run a Transient Analysis and plot the voltages.

b) Schematic Diagram and Simulation and Results

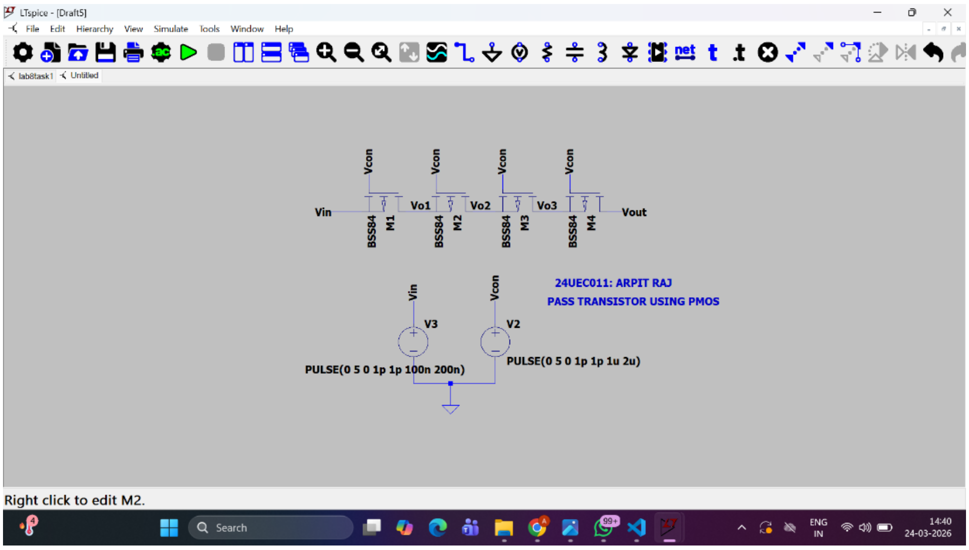

Figure 3: Schematic of Pass Transistor Using PMOS

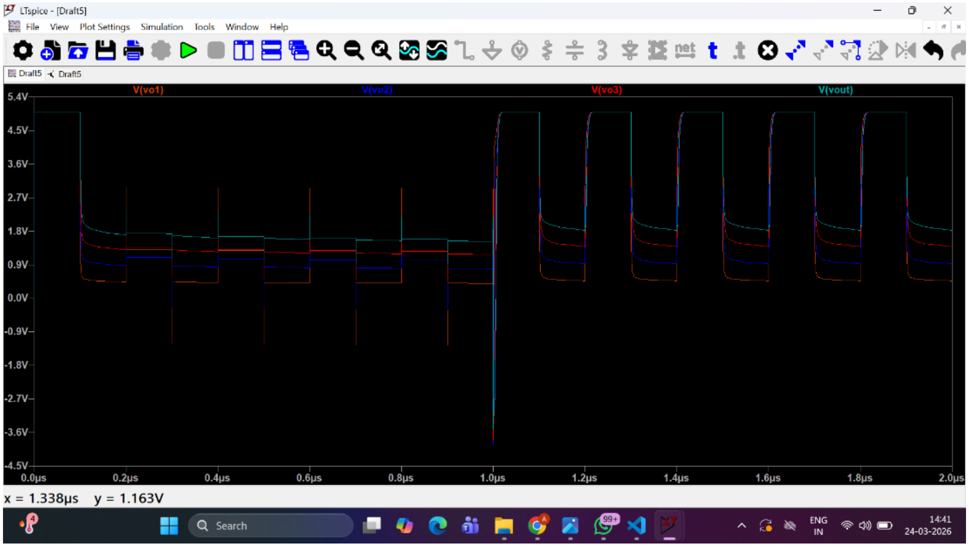

Figure 4: Simulation of Pass Transistor Using PMOS

0.4.3 2X1 MUX Using NMOS Pass Transistor Logic

a) Steps used to draw the Circuit

(a) An PTL based MUX works exactly like a CMOS.

(b) MUX but uses only 2 NMOS.

(c) When S = 1, NMOS Output is = B.

(d) When S = 0, NMOS Output is = A.

b) Schematic Diagram and Simulation and Results

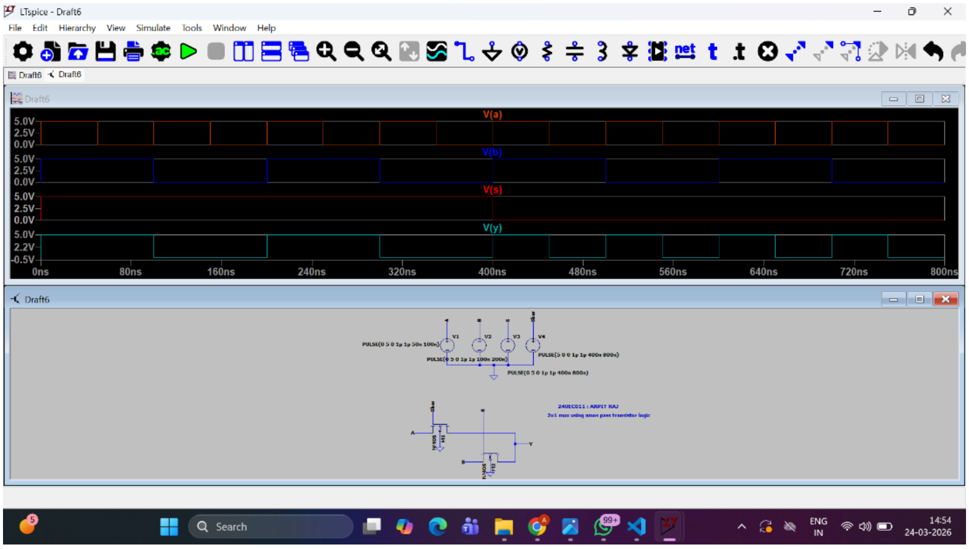

Figure 5: Schematic of 2X1 MUX Using NMOS Pass Transistor Logic

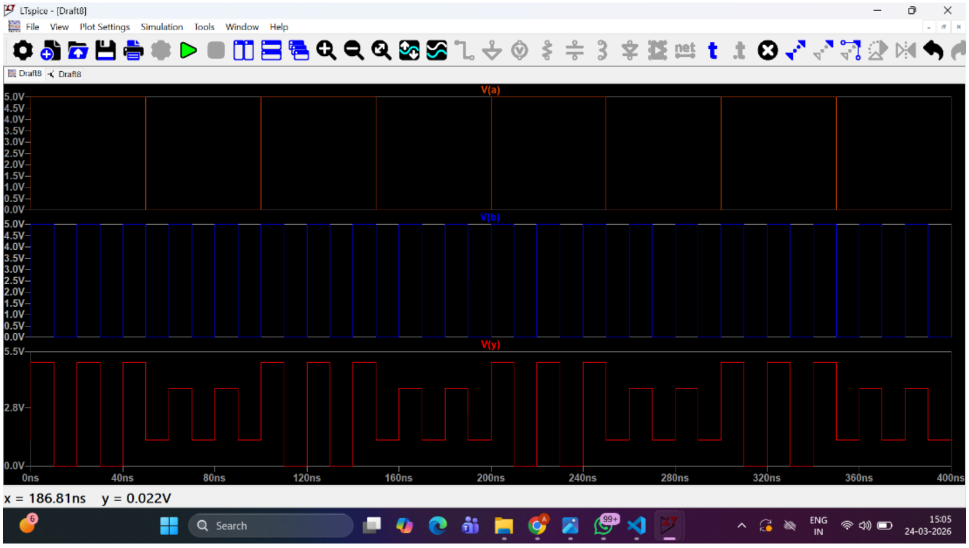

Figure 6: Simulation of 2X1 MUX Using NMOS Pass Transistor Logic

0.4.4 Transmission Gate Using NMOS and PMOS

a) Steps used to draw the Circuit

(a) A Transmission Gate (TG) uses NMOS and PMOS transistors in parallel.

(b) NMOS handles logic LOW (0V) well, while PMOS handles logic HIGH (5V) perfectly.

(c) Together, they achieve ideal signal transmission without voltage loss.

b) Schematic Diagram and Simulation and Results

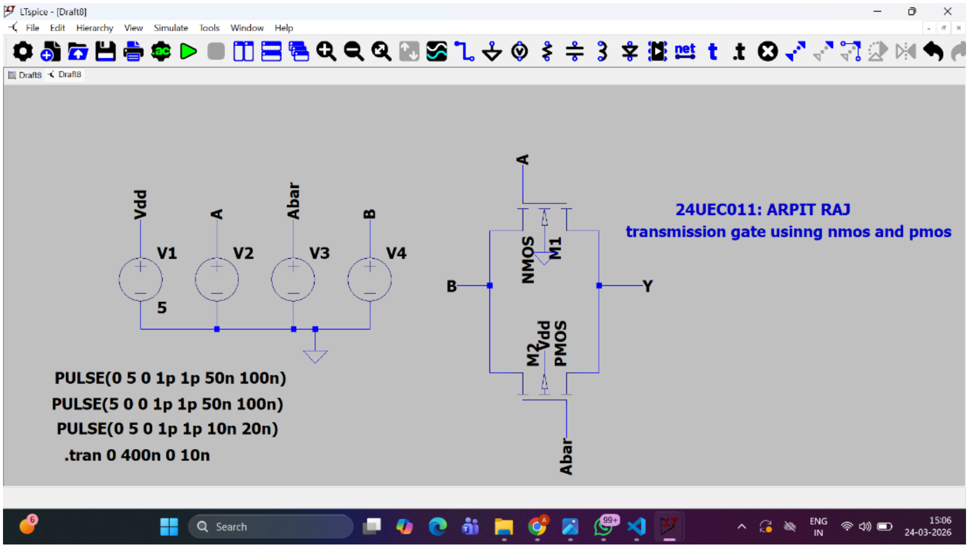

Figure 7: Schematic of Transmission Gate Using NMOS and PMOS

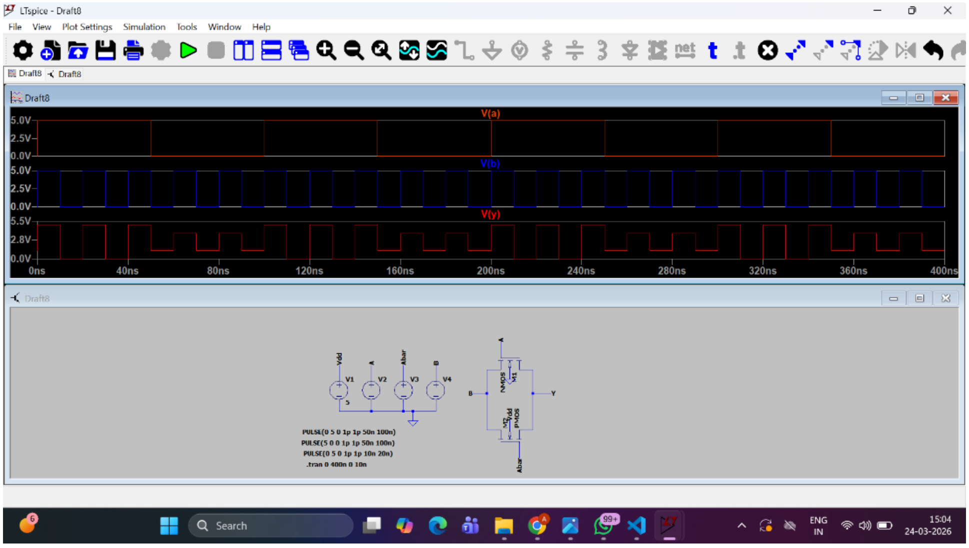

Figure 8: Simulation of Transmission Gate Using NMOS and PMOS

0.4.5 2x1 MUX using Transmission Gate Logic

a) Steps used to draw the Circuit

(a) A 2×1 multiplexer (MUX) is a fundamental combinational circuit that selects one of two input signals based on a control signal (SEL).

(b) The selected input is passed to the output.

(c) In Transmission Gate Logic (TGL), a 2×1 MUX is implemented using two transmission gates (TG), each controlled by the select signal (SEL) and its complement (SEL').

(d) This design ensures low power consumption and no voltage degradation, unlike pass transistor logic using NMOS or PMOS alone.

b) Schematic Diagram and Simulation and Results

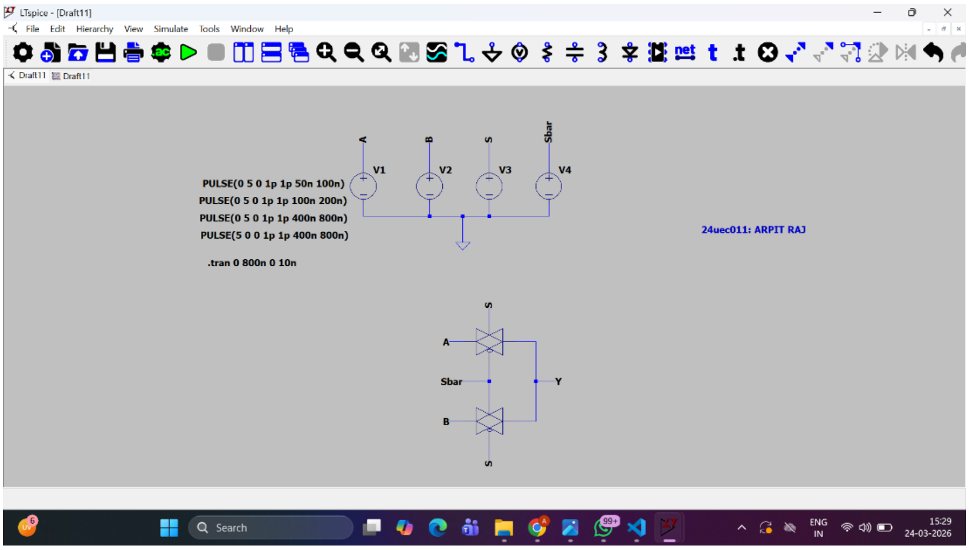

Figure 9: Schematic of 2x1 MUX using Transmission Gate Logic

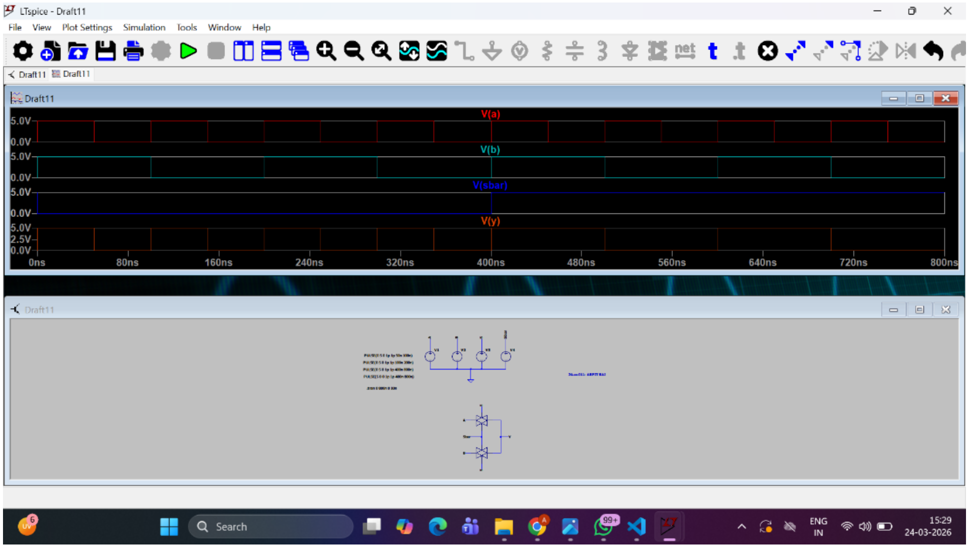

Figure 10: Simulation of 2x1 MUX using Transmission Gate Logic

0.5 Summary

- Successfully designed and simulated pass transistor circuits.
- Observed signal degradation in NMOS and perfect transmission.
- Demonstrated low power advantages of pass transistor logic.